MIT 6.1200J
Problem Set 3

Nicolas Iskandar

Problem 1

(a) **States**: The set of all permutations of $\{0, 1, \ldots, n-1\}$.

Start state: The initial permutation $(x_0, x_1, \ldots, x_{n-1})$.

Transitions: From a permutation $(a_0, \ldots, a_{n-1})$, there is a transition to

$$(a_0, \ldots, a_{i-1}, a_j, a_{j-1}, \ldots, a_{i+1}, a_i, a_{j+1}, \ldots, a_{n-1})$$

for every inverted pair (i, j) with $i < j$ and $a_i > a_j$ (i.e., only good reversals are allowed).

(b) **Final states**: A state has no outgoing transitions when no good reversal is possible, i.e., when there are no inverted pairs. A permutation has no inverted pairs if and only if it is sorted in increasing order. Therefore, the unique final state is the sorted permutation $(0, 1, 2, \ldots, n-1)$.

Partial correctness: If the machine terminates, it must be in a final state. Since the only final state is $(0, 1, \ldots, n-1)$, the list must be sorted. ■

(c) We define the following derived variable. Interpret the permutation $(a_0, a_1, \ldots, a_{n-1})$ as the digits of a base-n number:

$$V(a_0, \ldots, a_{n-1}) := \sum_{k=0}^{n-1} a_k \cdot n^{n-1-k}.$$

We claim V is a strictly decreasing derived variable with nonneg integer values.

Property (i) — nonneg integer values: Since each $a_k \in \{0, \ldots, n-1\}$ and $n^{n-1-k} \in \mathbb{N}$, we have $V \in \mathbb{N}$. ✓

Property (ii) — strictly decreasing: Suppose we perform a good reversal on the inverted pair (i, j), where $i < j$ and $a_i > a_j$. The new permutation $(a'_0, \ldots, a'_{n-1})$ satisfies:

- $a'_k = a_k$ for $k < i$ and $k > j$ (unchanged outside the reversed segment),
- the segment $(a'_i, \ldots, a'_j)$ is the reverse of $(a_i, \ldots, a_j)$, so in particular $a'_i = a_j$ and $a'_j = a_i$.

Since $a'_k = a_k$ for $k < i$ and $k > j$, only positions i through j contribute to $V' - V$:

$$V' - V = \sum_{k=i}^{j} (a'_k - a_k) \cdot n^{n-1-k}.$$

Each coefficient $a'_k - a_k$ is an integer, and each weight n^{n-1-k} for $k \leq j$ satisfies $n - 1 - k \geq n - 1 - j$, so $n^{n-1-j} \mid n^{n-1-k}$. Therefore every term is divisible by n^{n-1-j}, and so $V' - V$ is an integer multiple of n^{n-1-j}. It therefore suffices to show $V' - V < 0$: since $V' - V$ is an

integer multiple of n^{n-1-j}, a negative value must be at most $-n^{n-1-j}$, and $n^{n-1-j} \geq 1$ gives us $V' - V \leq -n^{n-1-j} \leq -1$.

To show $V' - V < 0$, factor out n^{n-1-j} from the difference:

$$\frac{V' - V}{n^{n-1-j}} = \sum_{k=i}^{j} (a'_k - a_k) \cdot n^{j-k}.$$

This is the difference between two base-n numbers formed from the same multiset of digits. In the new number, the most significant position $k = i$ has digit $a'_i = a_j$, while in the old it had digit $a_i > a_j$. Therefore the most significant digit has strictly decreased, which means the new number is strictly smaller than the old. Hence $V' - V < 0$, and therefore:

$$V' - V \leq -n^{n-1-j} \leq -1 < 0.$$

So V strictly decreases with every good reversal. ✓

(d) **Total correctness**: We combine parts (b) and (c).

Termination: By part (c), V is a derived variable that takes only nonneg integer values and strictly decreases with every transition. Since a nonneg integer cannot decrease forever, every execution must eventually terminate.

Partial correctness: By part (b), if the machine terminates, it must be in a final state, and the unique final state is the sorted permutation $(0, 1, \ldots, n-1)$.

Combining these two facts, every execution eventually terminates, and upon termination the list is fully sorted. ■

Problem 2

(a) We use the perturbation method. Let

$$S = \sum_{n=1}^{\infty} \frac{F_n}{c^n}.$$

Divide both sides by c:

$$\frac{S}{c} = \sum_{n=1}^{\infty} \frac{F_n}{c^{n+1}} = \sum_{n=2}^{\infty} \frac{F_{n-1}}{c^n}.$$

Similarly,

$$\frac{S}{c^2} = \sum_{n=1}^{\infty} \frac{F_n}{c^{n+2}} = \sum_{n=3}^{\infty} \frac{F_{n-2}}{c^n}.$$

Now compute $S - \frac{S}{c} - \frac{S}{c^2}$:

$$S - \frac{S}{c} - \frac{S}{c^2} = \sum_{n=1}^{\infty} \frac{F_n}{c^n} - \sum_{n=2}^{\infty} \frac{F_{n-1}}{c^n} - \sum_{n=3}^{\infty} \frac{F_{n-2}}{c^n}.$$

Separating the first two terms:

$$= \frac{F_1}{c} + \frac{F_2 - F_1}{c^2} + \sum_{n=3}^{\infty} \frac{F_n - F_{n-1} - F_{n-2}}{c^n}.$$

By the Fibonacci recurrence, $F_n - F_{n-1} - F_{n-2} = 0$ for all $n \geq 3$. Also $F_1 = 1$ and $F_2 = 1$, so:

$$S\left(1 - \frac{1}{c} - \frac{1}{c^2}\right) = \frac{1}{c} + \frac{0}{c^2} = \frac{1}{c}.$$

Therefore:

$$S = \frac{1/c}{1 - 1/c - 1/c^2} = \frac{1/c \cdot c^2}{c^2 - c - 1} = \frac{c}{c^2 - c - 1}.$$

Substituting $c = 10^5$:

$$\boxed{S = \frac{10^5}{10^{10} - 10^5 - 1} = \frac{100000}{9999899999}.}$$

(b) Using WolframAlpha to compute $S = \frac{100000}{9999899999}$ to 50 decimal places:

$$S \approx 0.\mathbf{00001}\ 00001\ 00002\ 00003\ 00005\ 00008\ 00013\ 00021\ 00034\ 00055\ldots$$

The pattern is clear: the digits after the decimal point, read in blocks of 5, give exactly the Fibonacci numbers $F_1, F_2, F_3, \ldots = 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$

This makes sense: by definition $S = \sum_{n=1}^{\infty} F_n/c^n$, so each Fibonacci number occupies its own block of 5 decimal digits (as long as $F_n < c = 10^5$, i.e., before carrying occurs).

Problem 3

Let $f(x) = \dfrac{x}{(5+x)^3}$.

(a) The integral method requires f to be eventually non-negative, continuous, and decreasing. While $f(x) > 0$ and is eventually decreasing, the issue is at $x = 0$: $f(0) = 0$. More critically, f is not monotonically decreasing starting from $x = 0$. We verify: $f'(x) = \frac{(5+x)^3 - 3x(5+x)^2}{(5+x)^6} = \frac{5-2x}{(5+x)^4}$, which is positive for $x < 5/2$ and negative for $x > 5/2$. So f is *increasing* on $[0, 5/2]$, meaning the integral method cannot be applied directly to S_0.

(b) We need f to be decreasing on $[i, \infty)$, which requires $i > 5/2$. Since i must be an integer, the smallest such index is $\boxed{i = 3}$.

(c) We apply the integral method to $S_3 = \sum_{x=3}^{\infty} f(x)$. Since f is decreasing on $[3, \infty)$:

$$\int_3^{\infty} f(x)\,dx \leq S_3 \leq f(3) + \int_3^{\infty} f(x)\,dx.$$

We compute the integral. Using the substitution $u = 5 + x$:

$$\int_3^{\infty} \frac{x}{(5+x)^3}\,dx = \int_8^{\infty} \frac{u-5}{u^3}\,du = \int_8^{\infty} \left(\frac{1}{u^2} - \frac{5}{u^3}\right)du = \left[-\frac{1}{u} + \frac{5}{2u^2}\right]_8^{\infty} = \frac{1}{8} - \frac{5}{128} = \frac{11}{128}.$$

Also, $f(3) = \frac{3}{8^3} = \frac{3}{512}$. So:

$$\frac{11}{128} \leq S_3 \leq \frac{11}{128} + \frac{3}{512} = \frac{44}{512} + \frac{3}{512} = \frac{47}{512}.$$

Now $S_0 = f(0) + f(1) + f(2) + S_3$. We compute:

$$f(0) = 0, \quad f(1) = \frac{1}{216}, \quad f(2) = \frac{2}{343}.$$

Let $C = f(0) + f(1) + f(2) = \frac{1}{216} + \frac{2}{343}$. Then:

$$\frac{11}{128} + C \leq S_0 \leq \frac{47}{512} + C.$$

The accuracy achieved is $\frac{47}{512} - \frac{11}{128} = \frac{47}{512} - \frac{44}{512} = \frac{3}{512} \approx 0.00586$.

(d) We want accuracy $\frac{1}{200}$, i.e., $f(j) \leq \frac{1}{200}$. Since the width of the interval from the integral method is $f(j)$ (the upper and lower bounds differ by exactly $f(j)$), we need the smallest integer $j \geq 3$ such that $f(j) \leq \frac{1}{200}$.

We check:

$$f(j) = \frac{j}{(5+j)^3} \leq \frac{1}{200}.$$

Testing $j = 7$: $f(7) = \frac{7}{12^3} = \frac{7}{1728} < \frac{1}{200}$ since $7 \cdot 200 = 1400 < 1728$. ✓

Testing $j = 6$: $f(6) = \frac{6}{11^3} = \frac{6}{1331}$. $\frac{6}{1331} \leq \frac{1}{200}$? We need $6 \cdot 200 \leq 1331$, i.e., $1200 \leq 1331$. ✓

Testing $j = 5$: $f(5) = \frac{5}{10^3} = \frac{5}{1000} = \frac{1}{200}$. We need $\frac{1}{200} \leq \frac{1}{200}$. ✓

Testing $j = 4$: $f(4) = \frac{4}{9^3} = \frac{4}{729}$. Is $\frac{4}{729} \leq \frac{1}{200}$? We need $4 \cdot 200 \leq 729$, i.e., $800 \leq 729$. ×

So the smallest such j is $\boxed{j = 5}$.

Now we apply the integral method to S_5:

$$\int_5^\infty \frac{x}{(5+x)^3}\,dx = \int_{10}^\infty \frac{u-5}{u^3}\,du = \left[-\frac{1}{u} + \frac{5}{2u^2}\right]_{10}^\infty = \frac{1}{10} - \frac{5}{200} = \frac{1}{10} - \frac{1}{40} = \frac{3}{40}.$$

So:

$$\frac{3}{40} \leq S_5 \leq \frac{3}{40} + f(5) = \frac{3}{40} + \frac{1}{200} = \frac{15}{200} + \frac{1}{200} = \frac{16}{200} = \frac{2}{25}.$$

Now $S_0 = f(0) + f(1) + f(2) + f(3) + f(4) + S_5$. Computing:

$$f(0) = 0, \quad f(1) = \frac{1}{216}, \quad f(2) = \frac{2}{343}, \quad f(3) = \frac{3}{512}, \quad f(4) = \frac{4}{729}.$$

Let $C' = \frac{1}{216} + \frac{2}{343} + \frac{3}{512} + \frac{4}{729}$. Then:

$$u = \frac{3}{40} + C' \leq S_0 \leq \frac{2}{25} + C' = v,$$

with $|v - u| = \frac{1}{200}$. ✓

Problem 4

(a) We write:

$$\sum_{i=0}^{n} \frac{5^i - 7^i}{13^i} = \sum_{i=0}^{n} \left(\frac{5}{13}\right)^i - \sum_{i=0}^{n} \left(\frac{7}{13}\right)^i.$$

Using the geometric series formula $\sum_{i=0}^{n} r^i = \frac{1-r^{n+1}}{1-r}$:

$$= \frac{1-(5/13)^{n+1}}{1-5/13} - \frac{1-(7/13)^{n+1}}{1-7/13} = \frac{13}{8}\left(1 - \frac{5^{n+1}}{13^{n+1}}\right) - \frac{13}{6}\left(1 - \frac{7^{n+1}}{13^{n+1}}\right).$$

Therefore:

$$\boxed{\sum_{i=0}^{n} \frac{5^i - 7^i}{13^i} = \frac{13}{8} - \frac{13}{6} + \frac{13}{6} \cdot \frac{7^{n+1}}{13^{n+1}} - \frac{13}{8} \cdot \frac{5^{n+1}}{13^{n+1}} = -\frac{13}{24} + \frac{7^{n+1}}{6 \cdot 13^n} - \frac{5^{n+1}}{8 \cdot 13^n}.}$$

(b) We evaluate the double sum by switching the order of summation:

$$\sum_{i=1}^{n} \sum_{j=1}^{k} (3j - i).$$

We split and evaluate each part separately:

$$\sum_{i=1}^{n} \sum_{j=1}^{k} (3j - i) = \sum_{i=1}^{n} \left(3 \sum_{j=1}^{k} j - \sum_{j=1}^{k} i\right) = \sum_{i=1}^{n} \left(\frac{3k(k+1)}{2} - ki\right).$$

Now summing over i:

$$= n \cdot \frac{3k(k+1)}{2} - k \sum_{i=1}^{n} i = \frac{3nk(k+1)}{2} - k \cdot \frac{n(n+1)}{2}.$$

Therefore:

$$\boxed{\sum_{i=1}^{n} \sum_{j=1}^{k} (3j - i) = \frac{nk}{2}\Big(3(k+1) - (n+1)\Big) = \frac{nk(3k - n + 2)}{2}.}$$